

ИССЛЕДОВАНИЕ

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Атрибуция на основе внимания: на что на самом деле смотрит ваша модель

Косинусное сходство — это косплей. Внимание — это атрибуция.



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В сфере генеративного искусственного интеллекта часто используются такие модные словечки, как «аудируемость», «объяснимость» и «интерпретируемость». Например, некоторые разработчики клинических систем утверждают, что предлагают нечто под названием *атрибуция*: для любого фрагмента текста, сгенерированного моделью, можно определить, какие фрагменты текста на входе оказали наибольшее «влияние» на эту часть генерации.

Однако определение влияния в этом контексте может быть весьма расплывчатым. Например, если компания (как в случае с вышеупомянутым автором) использует модели с закрытым исходным кодом, а все, что вы получаете от API модели, — это текст, то как вычислить степень влияния? По сути, у вас есть только один вариант: разбить текст на фрагменты, а затем сравнить сходство фрагментов в выходных данных с фрагментами во входных данных. Обычно это делается путем встраивания фрагментов (с помощью кодирующего трансформера, такого как `text-embedding-3-small` от OpenAI) и вычисления косинусного сходства между вложениями фрагментов.

Мы подчеркиваем, что это не влияние. Это полезная мера сходства, но она никак не показывает, что на самом деле делала модель, когда генерировала этот фрагмент. Таким образом, она не отражает и не может отразить реальные рассуждения модели. В частности, когда вы создаете некое количественное представление фрагмента текста (например, эмбединг), вы представляете только информацию,

содержащуюся в этом конкретном фрагменте, без учета контекста предшествующей и последующей информации. Ниже мы рассмотрим конкретный пример, в котором подход с использованием косинусного сходства не работает, но в целом можно сказать следующее: не стоит доверять компании, которая использует только API-интерфейсы моделей с закрытым исходным кодом, даже если она утверждает, что обеспечивает атрибуцию и возможность аудита. На самом деле она предоставляет вам наспех собранную произвольную меру сходства, которая не связана с реальной моделью, выполняющей генерацию. Для обеспечения реальной объяснимости вам нужен доступ к *внутреннему устройству модели*. К счастью, поскольку в Baseten мы дорабатываем модели с открытым исходным кодом, мы можем это сделать.

Где косинусное сходство терпит неудачу

Рассмотрим простую задачу: модели дают расшифровку аудиозаписи и просят составить клиническое заключение, как это сделал бы врач. Мы видим, что в приведенном ниже примере модель составила список из трех травм: ссадина на лодыжке, рваная рана на запястье и ушиб лба. Атрибуция очень полезна для визуализации того, откуда модель могла взять информацию для создания той или иной части вывода. Мы можем сделать это, например, чтобы проверить, не сгенерировала ли модель какую-то часть изображения случайным образом, а не на основе информации из входных данных.

Для начала давайте рассмотрим ошибочный подход с использованием косинусного сходства на этом синтетическом примере. Откуда, по его мнению, взялась эта информация? Напомним, что при таком подходе, чтобы получить оценку атрибуции для этого конкретного «фрагмента» выходных данных, мы просто встраиваем выходной фрагмент, встраиваем все входные фрагменты и вычисляем косинусное сходство между каждой парой входных и выходных фрагментов. (В этом примере мы разбиваем текст только на новые строки и точки.) Вот как в данном случае выглядит атрибуция (то есть три верхних фрагмента, связанных с выбранным выходным фрагментом):

TRANSCRIPT

Doctor: Hi Ms. Johnson, I understand you had a fall yesterday? Tell me what happened.

Patient: Yes, I was coming down the stairs at home and I slipped. I caught myself on the railing but still went down pretty hard.

Cosine: 0.567 I'm sorry to hear that. Where did you hurt yourself?

Patient: My left ankle twisted when I fell, and I scraped my right wrist trying to catch myself. Oh, and I hit my forehead on the railing.

Doctor: Okay, let's take a look at each of those. Let me start with your ankle - can you describe what it looks like now?

Patient: It's swollen and there's a scrape on the outside part, maybe two inches long. It was bleeding **Cosine: 0.503** but I cleaned it.

Cosine: 0.521 Good. And your wrist injury?

Patient: The wrist has a deeper cut, probably an inch long on the back of my hand near the wrist. It bled quite a bit initially.

Doctor: Did you clean that one too?

Patient: Yes, I used soap and water and put a bandage on it.

Doctor: And the forehead injury?

Patient: It's a bump with a small cut, maybe half an inch. Above my right eyebrow.

Doctor: Alright. I'm looking at your chart and I see you have diabetes - are you still taking metformin?

Patient: Yes, 1000 milligrams twice a day.

Doctor: And how have your blood sugars been lately?

Patient: Honestly, not great. Usually around 200 to 250. I've been trying to do better with my diet.

Doctor: I appreciate you being honest about that. That's important for wound healing. Let me examine these injuries now. Looking at your ankle - the scrape is superficial, but there's some redness around it. Does it hurt?

Patient: Yes, it's tender to touch. More than I expected.

Doctor: And when you walk on it?

Patient: It hurts, but I can put weight on it.

Doctor: Okay. Now your wrist - this cut is deeper than you described. I can see it goes into the dermis. Is it oozing at all?

Patient: A little bit, yes. I noticed some yellow stuff on the bandage this morning.

Doctor: That concerns me a bit. And you're also on blood pressure medication, correct? I see lisinopril in your chart.

Patient: Yes, 10 milligrams once a day in the morning.

Doctor: Good to know. Let me check your forehead now. This wound is small but it's in a spot where we need to watch for infection carefully. When did you last clean it?

Patient: This morning, same as the others.

Doctor: The ankle - you mentioned redness. Has that gotten worse or stayed the same?

CLINICAL NOTE

****CHIEF COMPLAINT:**** Multiple injuries sustained from fall down stairs.

****HISTORY OF PRESENT ILLNESS:**** Patient is a diabetic female who presents one day after falling down stairs at home. She caught herself on railing while falling. **Sustained three injuries: (1) Left lateral ankle with 2-inch superficial abrasion, (2) Right dorsal wrist with 1-inch laceration extending into dermis, and (3) Small 0.5-inch laceration with contusion to right forehead above eyebrow.** Patient cleaned all wounds with soap and water and applied bandages. She reports increased redness around ankle wound since yesterday with spreading erythema. Wrist wound has purulent drainage noted on dressing this morning. Forehead wound cleaned this morning without reported complications.

****PAST MEDICAL HISTORY:**** Type 2 diabetes mellitus with poor glycemic control (recent blood glucose 200-250 mg/dL).

****MEDICATIONS:**** Metformin 1000mg PO BID, Lisinopril 10mg PO daily.

****ALLERGIES:**** No known drug allergies.

****PHYSICAL EXAMINATION:****

- Left ankle: 2-inch superficial abrasion on lateral aspect with surrounding erythema and tenderness. Wound is dry. Patient ambulates with pain but able to bear weight.
- Right wrist: 1-inch deep laceration on dorsal hand/wrist with purulent drainage. Extends into dermal layer.
- Forehead: 0.5-inch laceration with contusion, right supraorbital region. No active drainage.

****ASSESSMENT:**** Multiple traumatic wounds with concern for infection secondary to poor diabetes control. Impaired wound healing risk given persistent hyperglycemia. Spreading erythema at ankle site and purulent drainage at wrist site suggest early wound infection.

****PLAN:****

1. Initiate oral antibiotics for wound infection coverage
2. Daily wound care with cleaning and dressing changes
3. Close monitoring for progression of erythema or drainage
4. Reinforce importance of glycemic control for wound healing
5. Return in 48 hours for wound check or sooner if worsening symptoms
6. Consider wound culture if no improvement with empiric antibiotics

END CLINICAL NOTE

Attribution Strength

- Target Output
- Top Attribution
- Medium Attribution
- Low Attribution

Косинусное сходство полностью игнорирует третью рану (на лбу)! Понятно, почему косинусное сходство не учитывает эту информацию. Поскольку эмбединг для конкретного фрагмента может содержать только информацию, относящуюся к этому фрагменту, у него нет возможности понять, что третья рана относится ко лбу, о котором говорится в начале расшифровки («А рана на лбу?»).

Теперь давайте посмотрим, работает ли атрибуция на основе внимания лучше. Здесь мы выбираем один слой предварительно обученного трансформера **qwen2.5-72b** (слой 40) и рассчитываем тепловую карту внимания, усредняя значения внимания по головам в слое и суммируя значения внимания по фрагментам.

TRANSCRIPT

****Doctor:**** Hi Ms. Johnson, I understand you had a fall yesterday? Tell me what happened.

****Patient:**** Yes, I was coming down the stairs at home and I slipped. I caught myself on the railing but still went down pretty hard.

Attention: 0.090 ****Patient:**** I'm sorry to hear that. Where did you hurt yourself?

****Patient:**** My left ankle twisted when I fell, and I scraped my right wrist trying to catch myself. Oh, and I hit my forehead on the railing.

****Doctor:**** Okay, let's take a look at each of those. Let me start with your ankle - can you describe what it looks like now?

****Patient:**** It's swollen and there's a scrape on the outside part, maybe two inches long. It was bleeding yesterday but I cleaned it.

Attention: 0.080 ****Patient:**** And your wrist injury?

****Patient:**** The wrist has a deeper cut, probably an inch long on the back of my hand near the wrist. It bled quite a bit initially.

****Doctor:**** Did you clean that one too?

****Patient:**** Yes, I used soap and water and put a bandage on it.

Attention: 0.070 ****Patient:**** I the forehead injury?

****Patient:**** It's a bump with a small cut, maybe half an inch. Above my right eyebrow.

****Doctor:**** Alright. I'm looking at your chart and I see you have diabetes - are you still taking metformin?

****Patient:**** Yes, 1000 milligrams twice a day.

****Doctor:**** And how have your blood sugars been lately?

****Patient:**** Honestly, not great. Usually around 200 to 250. I've been trying to do better with my diet.

****Doctor:**** I appreciate you being honest about that. That's important for wound healing. Let me examine these injuries now. Looking at your ankle - the scrape is superficial, but there's some redness around it. Does it hurt?

****Patient:**** Yes, it's tender to touch. More than I expected.

****Doctor:**** And when you walk on it?

****Patient:**** It hurts, but I can put weight on it.

****Doctor:**** Okay. Now your wrist - this cut is deeper than you described. I can see it goes into the dermis. Is it oozing at all?

****Patient:**** A little bit, yes. I noticed some yellow stuff on the bandage this morning.

****Doctor:**** That concerns me a bit. And you're also on blood pressure medication, correct? I see lisinopril in your chart.

****Patient:**** Yes, 10 milligrams once a day in the morning.

****Doctor:**** Good to know. Let me check your forehead now. This wound is small but it's in a spot where we need to watch for infection carefully. When did you last clean it?

****Patient:**** This morning, same as the others.

****Doctor:**** The ankle - you mentioned redness. Has that gotten worse or stayed the same?

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- Left ankle: 2-inch superficial abrasion on lateral aspect with surrounding erythema and tenderness. Wound is dry. Patient ambulates with pain but able to bear weight.
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****ASSESSMENT:**** Multiple traumatic wounds with concern for infection secondary to poor diabetes control. Impaired wound healing risk given persistent hyperglycemia. Spreading erythema at ankle site and purulent drainage at wrist site suggest early wound infection.

****PLAN:****

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5. Return in 48 hours for wound check or sooner if worsening symptoms
6. Consider wound culture if no improvement with empiric antibiotics

END CLINICAL NOTE

Attribution Strength

- Target Output
- Top Attribution
- Medium Attribution
- Low Attribution

Attention gets it! It understands that "It's a bump with a small cut" refers to the forehead injury, because the attention information is not just localized to that particular chunk - by definition, attention uses the context of everything that has come before it to figure out where to look. This is a very simple and contrived example, but hopefully it gives some intuition of why cosine similarity often breaks down, and where using model internals like attention can improve on it.

Developing attention attribution into a useful tool

Of course, once you understand how attention works, it's the most obvious thing in the world that we'd want to use it to do attribution. However, there are still a lot of choices we need to make. First, we must decide whether to extract attention from the actual generating model (providing ground-truth attribution but requiring significant computational overhead) or from a smaller proxy model (trading some accuracy for efficiency). If using a proxy model, we need to determine: Which model size provides sufficient attention fidelity? Which specific layers capture the most relevant attention patterns? And crucially, what's the minimal subset of layers that maintains high correlation with full-model attention while remaining computationally feasible?

We also have some real-world engineering constraints. In a live inference setting, attribution must add negligible latency (< 100ms) to maintain responsive user experience—ruling out complex post-processing like iterative attention sink normalization that can

take seconds on long contexts (although we still do some form of attention sink normalization). More critically, during generation, our GPUs operate at near-maximum memory utilization to maximize throughput; storing full attention tensors for a 70B parameter model with 80 layers would require ~10-50GB of additional memory per request (depending on sequence length), which would force us to reduce batch sizes and directly impact serving capacity. These constraints drove our architectural decision: we decouple attribution from generation entirely, running a dedicated attribution model on separate, smaller GPUs. This separation allows us to optimize each system independently: the generation cluster for maximum throughput, and the attribution system for memory-efficient attention extraction using only the most informative layers (typically 10-15 layers rather than all 80), reducing memory requirements by 5-8x while maintaining >0.9 correlation with full-model attention patterns.

Establishing a ground-truth

We began by approaching the problem practically. Unfortunately, you can't just use all the attention layers from a model to get something meaningful, because different layers focus on different things. Much like how the layers in CNNs focus on increasingly abstract concepts (looking at low-level things like colors and lines in early layers, onto more complex patterns in the middle layers through to abstract concepts like “dog ear” in the later layers), transformer attention patterns follow a similar ascending complexity. Earlier attention layers tend to focus on grammatical structures, whereas later layers start to look at semantically related information relevant to the current token it's attending to. Simply averaging across all layers would conflate these distinct attention types, obscuring the semantic attribution signal with syntactic noise from early layers and over-specialized task features from final layers.

We did this in a pretty practical way. We got a team of doctors (led by our CEO and ex-doctor, Mudi) to do blind A/B tests with different model layers and subsets of layers, to see which points maximized the “intuitive agreement” with the model attention. We found that layers from about halfway through the model up until a couple of layers before the final layer worked best and provided what seemed to be the most human-understandable view of where the model was looking. These experiments were done on both **gemma-3-27b** and **qwen2.5-32b** models. Although not entirely principled, we choose layers 38-56 in the **qwen2.5-32b** model as the ground-truth attention source (averaging over both layers and attention heads for this set of layers). All correlations and comparisons are based on this ground truth attention. We discuss more holistic derivations of how to get out the full attention information in the appendix, and the drawbacks of doing so.

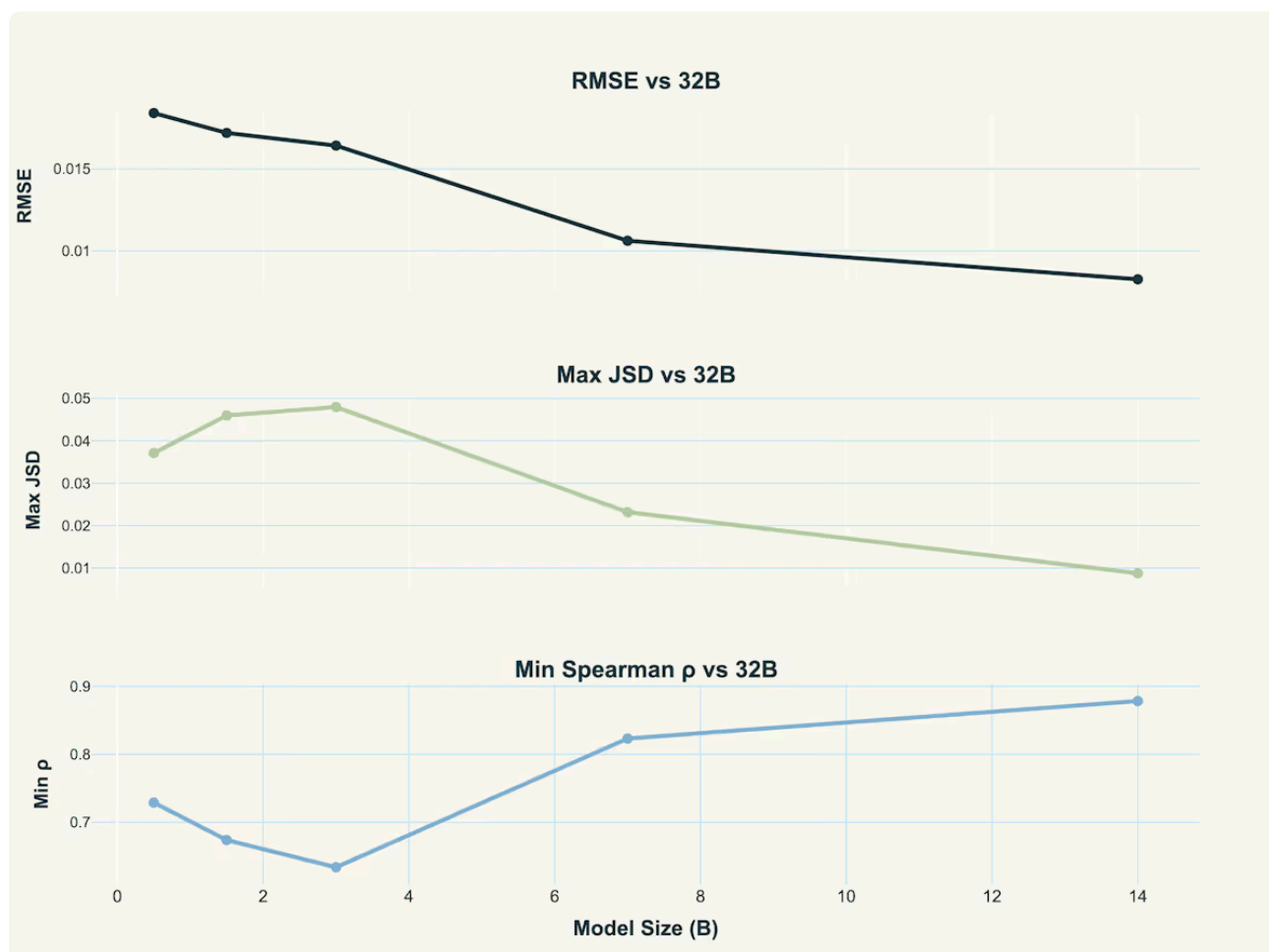
Measuring Chunk-Level Attention Similarity

We employ three complementary metrics to quantify how well a smaller model's chunk-to-chunk attention patterns match the 32B reference model. RMSE provides an overall

element-wise distance measure, treating the attention matrix as a whole and computing the L2 norm of differences. Maximum Jensen-Shannon Divergence (Max JSD) evaluates the worst-case distributional difference across query rows, treating each target chunk's attention distribution over source chunks as a probability distribution and computing the symmetric KL divergence, then taking the maximum across all rows to identify the most problematic target chunk. Minimum Spearman ρ (Min Spearman ρ) captures rank-order correlation by computing the Spearman correlation coefficient for each row independently (testing whether chunks attend to sources in the same relative order) and reporting the minimum to identify where ranking breaks down most severely.

Correlation across model sizes

The first thing we want to look at is whether smaller models can approximate the attention of the larger, ground-truth model (say, the model doing the generating). To do this, we take the same relative slice of layers (i.e. layers 0.6 to 0.9 of the way through the model) and average the attention across heads and layers. Interestingly, we can get quite close to **qwen2.5-32b** with the 14B and even 7B model.



It's also interesting that there is a drop off in similarity from 0.5B to 1.5B to 3B. This is possibly because the fractional relative slice of layers is not well calibrated for these

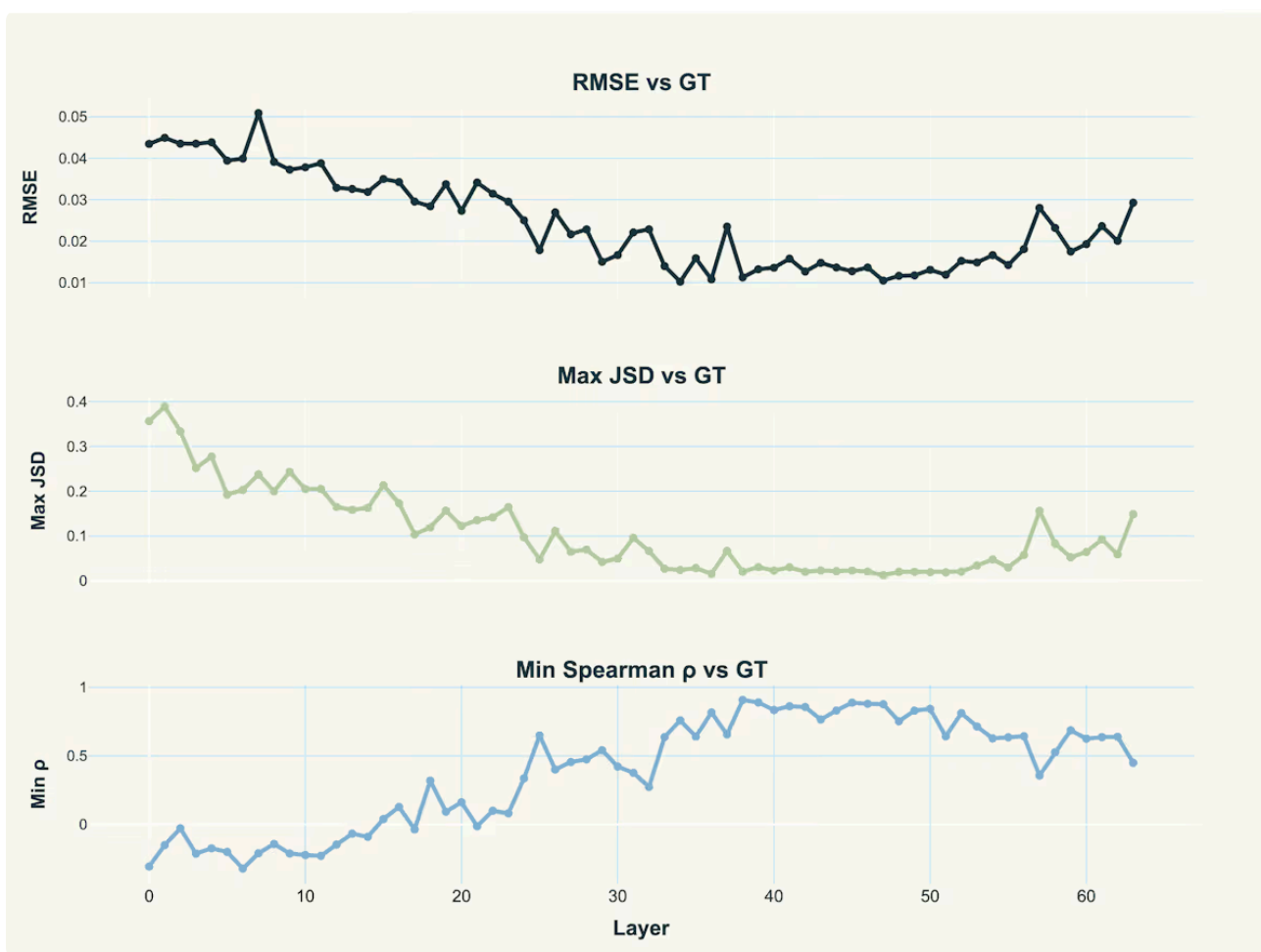
models, but also because the qwen models are weird in that they don't monotonically increase in depth as you go up the model family.

Regardless, most of the middle to later layers in the semi-large models have good correlation with the ground-truth from the 32B model.



Correlation across subsets of model layers

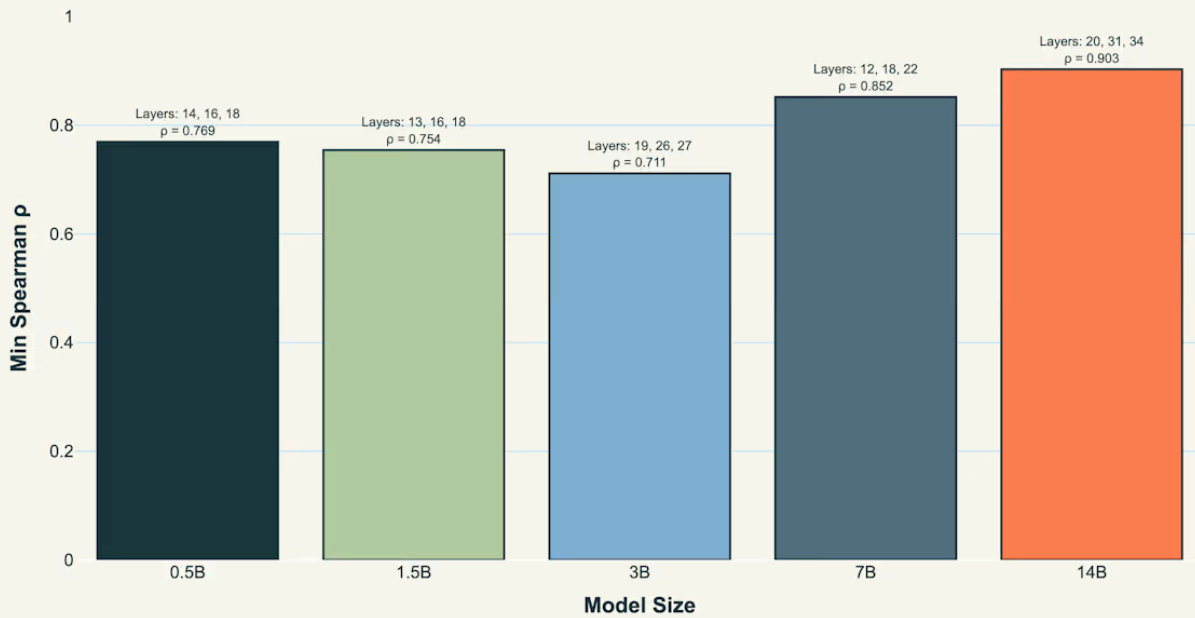
Of course, another way to trade off memory and compute is by only using a subset of layers from the same model (i.e. from `qwen2.5-32b`). To look at which are the best layers to pick, we can calculate the same statistics by comparing the attention patterns of an individual layer to the “ground-truth” aggregated attention patterns.



Clearly, most layers in the range we actually picked (38-56) correlate quite well with the averaged attention, suggesting it doesn't vary that much. This is great! It means we can get away with a subset of layers, possibly even as few as 1 to 3.

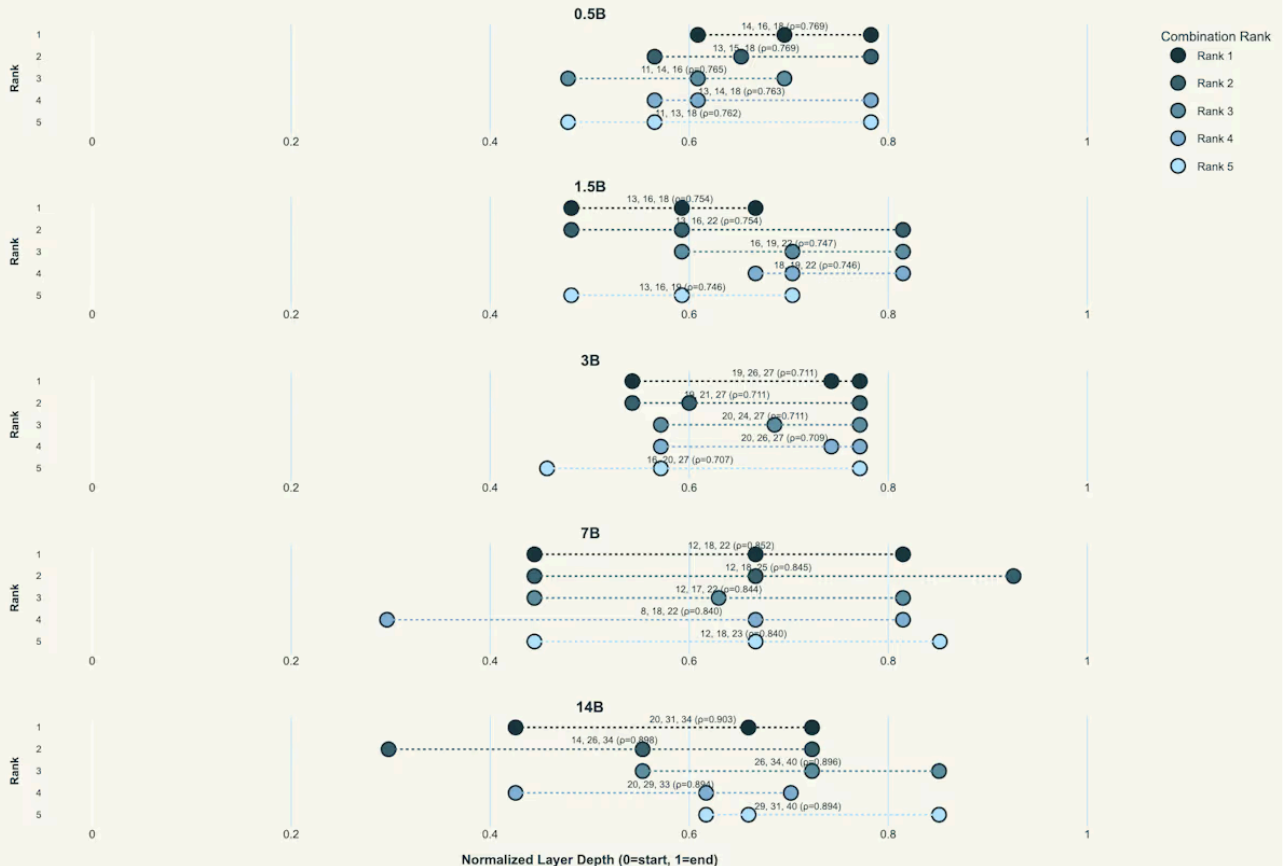
Let's say we have enough compute on the instance to be able to do 3 layers comfortably. We can also do a search over all the permutations of layers that lead to the highest correlation, across models.

Best 3-Layer Combinations: Min Spearman ρ vs 32B Ground Truth



A better way to visualize this is to show the distribution of the top 5 layer choices, across the models. We again normalize the layer depth. The spread is roughly the same - about halfway through the model, roughly evenly spaced, ending about four-fifths through the layers.

Distribution of Best 3-Layer Combinations Across Model Sizes



In practice, we use a collection of these automated analyses and intuitive heuristics when we are attributing with a new model to choose the best subset of layers and best model to approximate the “ground-truth” attention. However, important to note is that some customers *need* the ground-truth for regulation or compliance purposes, in which case we use the actual model that does the generation and essentially the full layers end-to-end (see Appendix for a discussion of this).

Attention attribution in Baseten Training

The best part of all this is that it's live in the Baseten Training platform! Our customers not only get access to fine-tuned models optimized for their task, but they can also get this attention-based attribution information. We usually send this back via our API, but here we show visually what attribution looks like in the *Playground*. Importantly, we can also take outputs from any closed-source model and use attention to generate attributions for it (which is useful here where we can't publicly display an individual customer's model, but in general allows us to perform attribution on *arbitrary* pieces of text):

--Left ankle:** Swollen, superficial abrasion (~2 inches), increasing redness, tender to touch, no drainage. Hurts to walk but weight bearing is possible.

--Right wrist/hand:** Deep laceration (~1 inch) on dorsum near wrist, initial heavy bleeding, now persistent mild oozing (yellow drainage), tenderness.

--Forehead:** Small abrasion (~0.5 inch) with bump above right eyebrow, cleaned this morning.

- Denies fever or other systemic symptoms.

- No known drug allergies.

Diabetes Control: Glucose levels running 200–250 mg/dL, dietary efforts ongoing but suboptimal control.

Medications:

- Metformin 1000 mg BiD
- Lisinopril 10 mg QAM

Objective:

--Vitals:** [Insert if known]

--General:** Alert, in no acute distress.

--Left Ankle:** Mild-moderate swelling, superficial abrasion (2"), surrounding erythema with mild expansion according to patient, tender. No open deep wound, no drainage.

--Right wrist/hand:** Laceration (~1" long), appears to reach dermis, mild yellowish exudate/oozing, surrounding skin mildly erythematous.

--Forehead (above R eyebrow):** Small abrasion (0.5"), mild swelling, no gross signs of infection.

--Gait:** Able to bear weight, cautious ambulation.

--Other:** No other injuries noted.

Assessment:

1. Fall with multiple minor injuries** (left ankle abrasion, right wrist laceration, superficial forehead abrasion).
2. Poorly controlled type 2 diabetes** increased risk for poor wound healing and infection.

Doctor: I appreciate you being honest about that. That's important for wound healing. Let me examine these injuries now. Looking at your ankle - the scrape is superficial, but there's some redness around it. Does it hurt?

Patient: Yes, it's tender to touch. More than I expected.

Doctor: And when you walk on it?

Patient: It hurts, but I can put weight on it.

Doctor: Okay. Now your wrist - this cut is deeper than you described. I can see it goes into the dermis. Is it oozing at all?

Patient: A little bit, yes. I noticed some yellow stuff on the bandage this morning.

Doctor: That concerns me a bit. And you're also on blood pressure medication, correct? I see lisinopril in your chart.

Patient: Yes, 10 milligrams once a day in the morning.

Doctor: Good to know. Let me check your forehead now. This wound is small but it's in a spot where we need to watch for infection carefully. When did you last clean it?

Patient: This morning, same as the others.

Doctor: The ankle - you mentioned redness. Has that gotten worse or stayed the same?

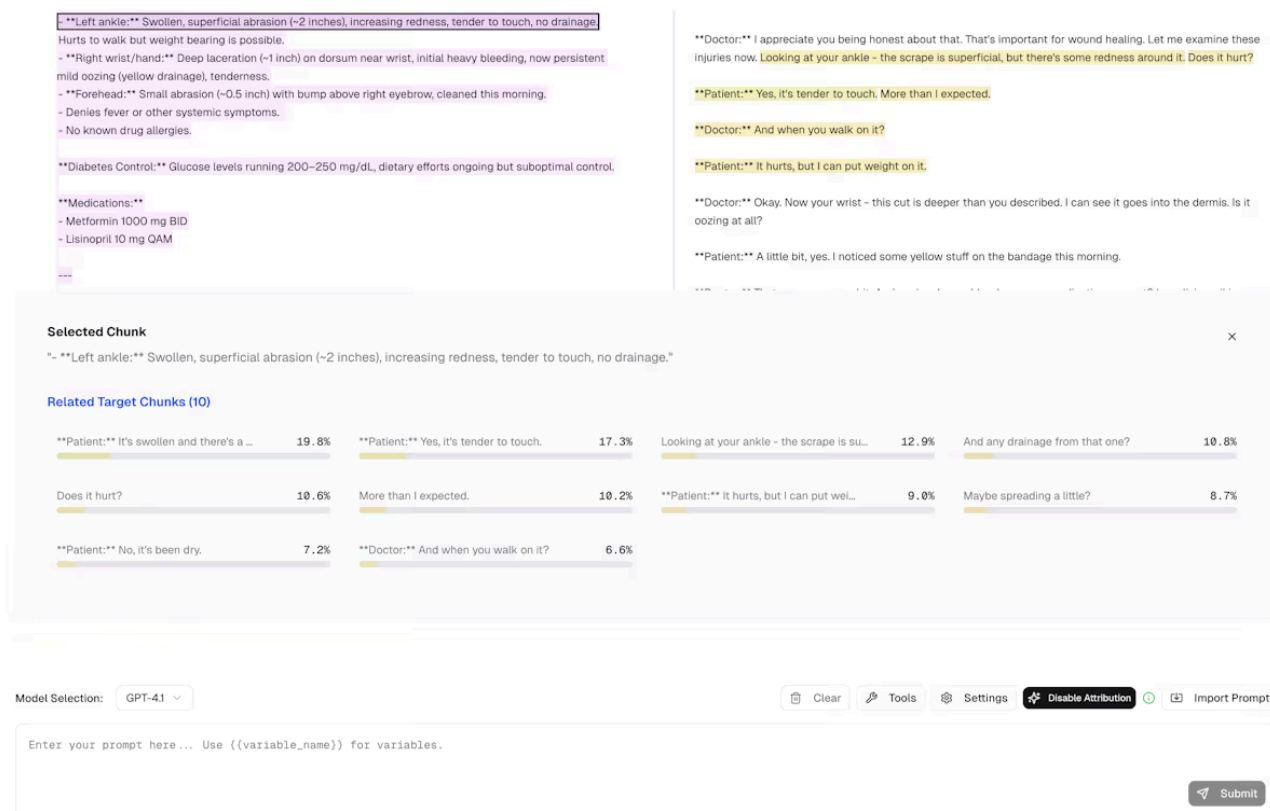
Patient: Actually, I think it's gotten a bit redder since yesterday. Maybe spreading a little?

Doctor: That's what I wanted to check. And any drainage from that one?

Patient: No, it's been dry.

Doctor: Alright. Do you have any drug allergies I should know about?

We can also click on any particular output chunk, and see the related target chunks in the input in a popup. Clicking on any of these related chunks will take you to the point in the input where that target chunk is located.



Our customers use this in a variety of circumstances:

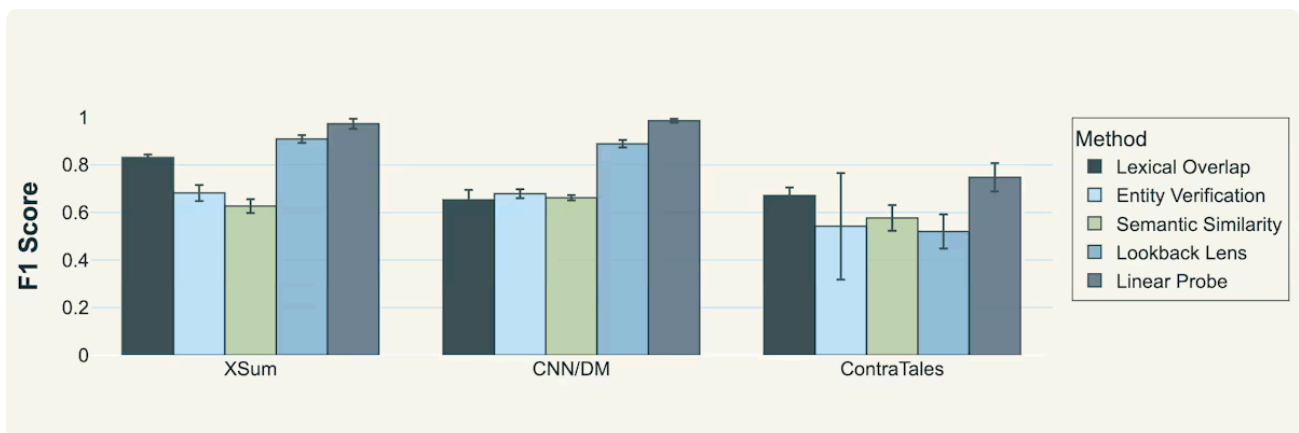
- Internal engineering teams use it to quickly sanity-check the outputs of models
- Quality-assurance/internal annotation and labelling teams (such as the medical knowledge teams in our scribe customers) use it as a very efficient way of seeing if the model has used the information in the input correctly, and whether there are parts of the output that are hallucinated (ie not derived from) the input
- Passing through the attribution information returned in our API to end-users and building their own frontends around it. For instance, some of our scribe customers have built the attribution hover-and-click mechanisms into their platforms so that doctors themselves can quickly scan and see where the model was looking for particular parts of the notes.

Of course, such ground-truth auditability is useful in a large number of domains. Healthcare (particularly scribes) are obviously useful places to apply this, but another place where interpretability has found a home is in the insurance industry. Here, it's so important that the model only use information from provided policy documents, for instance, and not its own knowledge. Attention attribution is a much more fail-safe way to check for this, as opposed to LLM-as-judge or cosine-similarity-based attribution.

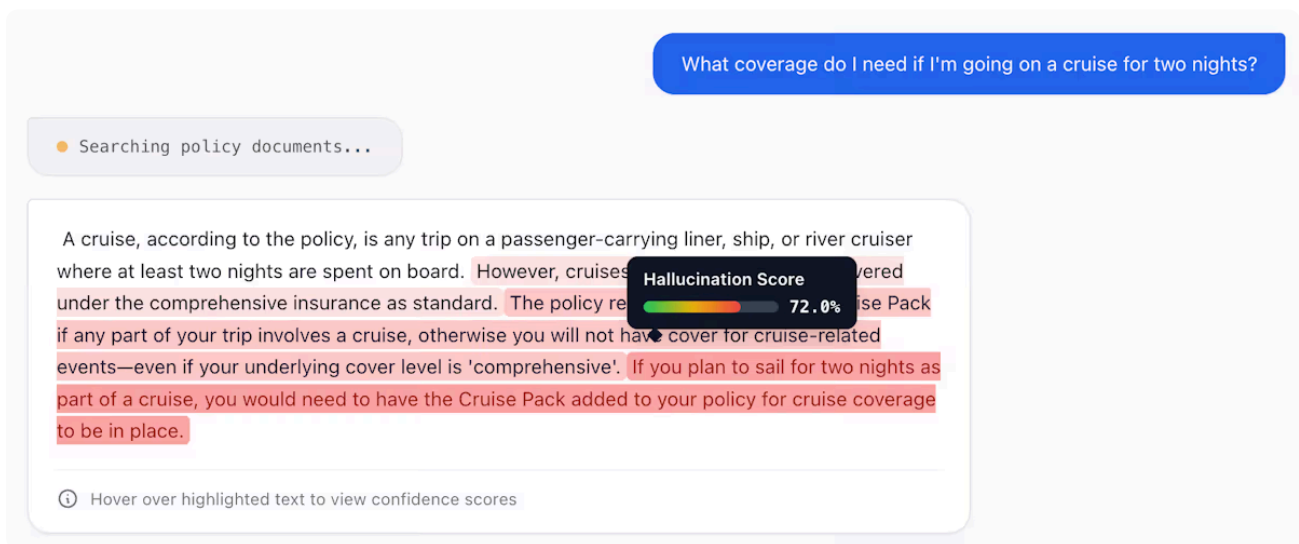
Future work

One might imagine the garden of useful mechanistic interpretability techniques to be a ladder of increasing complexity. Attention attribution is quite simple and is thus quite a low rung on this ladder. Motivated by using attention (or any model internals) to diagnose hallucinations at a chunk-level in model outputs, we asked ourselves: what if we used model internals to try and predict if a particular chunk is hallucinated (i.e. not supported by the input context) or not?

We initially thought we might be able to do this by training lightweight classifiers over the attention matrices themselves (an extension of ideas such as [Lookback Lens](#)) but found that training these same linear probes on the *residual stream* of the transformer (where the model “stores” all its intermediate computations) worked best. We published [a paper](#) on this at ICML, where we showed that not only can we, with high accuracy, predict whether a particular generated chunk was hallucinated by using the same model that generated it, but that we could actually take any model’s generation and use a fixed “judge” model’s internal activations to predict whether a particular chunk was hallucinated (i.e. whether it was supported by preceding input or not). This simple method outperformed all other approaches we tried, including semantic similarity (cosine-similarity), lexical overlap, entity verification and lookback lens.



This allows us to do cool stuff like flagging hallucinations live as we send back responses to customers. Whilst this is only available in the API currently, one might imagine being able to display this information in a similar manner to how we display attribution scores.



Another useful direction we've been looking into is trying to answer the following question. Suppose we have a model which has been fine-tuned to be really good at a particular task (say generating clinical notes). What is the difference between the attention this specially tuned model pays to parts of the input, and the attention of the base (not fine-tuned) version of the model? If the attention is exactly the same, then we can serve our attention attribution more efficiently by using one base model deployment for all customers using attention attribution. Our initial hypothesis is that they're quite similar, supported by evidence such as the fact that fine-tuning LoRA adapters on attention components only does not really increase performance, so a lot of the updates are probably confined to the MLPs. This is something we're actively investigating.

Finally, we are also looking at methods which better approximate the full model attention across layers, taking into account the mixing of information from different tokens. We discuss this in the Appendix.

Appendix

A more principled derivation

While individual attention matrices show local token-to-token relationships at each layer, they don't capture the full picture of how information propagates through the network's depth. This appendix derives a principled method for computing the **total effective attention** from input to output, accounting for both attention mechanisms and residual connections.

The model

At each layer ℓ , the transformer computes:

$$\mathbf{h}^{\ell+1} = \mathbf{h}^{\ell} + \text{Attention}^{\ell}(\mathbf{h}^{\ell}) + \text{MLP}^{\ell}(\mathbf{h}^{\ell})$$

For attention-only analysis (ignoring MLPs), we can linearize this as:

$$\mathbf{h}^{\ell+1} = (\mathbf{I} + \mathbf{A}^{\ell}) \mathbf{h}^{\ell}$$

where $\mathbf{A}^{\ell}[i, j]$ is the attention weight from token i to token j at layer ℓ , and $\mathbf{I}$ is the identity matrix representing the residual connection.

To understand how information flows from input ($\mathbf{h}^0$) to layer L , we compose these transformations:

$$\mathbf{h}^L = (\mathbf{I} + \mathbf{A}^L)(\mathbf{I} + \mathbf{A}^{L-1}) \cdots (\mathbf{I} + \mathbf{A}^1) \mathbf{h}^0$$

Expanding this product reveals how information propagates through multiple paths:

- **Direct paths:** Information flowing straight through residual connections $\mathbf{I} \cdots \mathbf{I}$
- **Single-hop paths:** Information modified by attention at exactly one layer
- **Multi-hop paths:** Information transformed by attention at multiple layers

The effective attention matrix that captures all information from input to output is:

$$\mathbf{A}_{\text{eff}} = \left(\prod_{\ell=1}^L (\mathbf{I} + \mathbf{A}^{\ell}) \right) - \mathbf{I}$$

Subtracting $\mathbf{I}$ removes the direct residual path, leaving only the attention-mediated influence.

However, we find that this doesn't necessarily provide the most useful attention patterns. What ends up happening is that everything converges to a small set of input chunks, and there is not a huge amount of variance in these input chunks. This is probably because earlier layers again have high attention on grammatical tokens ("the", ",", etc) and this converges to a pretty sparse attention pattern. We can run this process in theory on just the layers we care about (i.e. the layers we chose above to average for `qwen2.5-32b`) and this is something we're looking into as to whether this ends up better.

(Upon deriving the above, we realised this had been done before in [this paper](#).)

What exactly should $\mathbf{A}^{\ell}$ be?

Here, we set A^ℓ to raw attention probabilities (averaged over heads), and the product is thus the classic attention rollout heuristic. It captures routing structure, but not the magnitude effects of value and output projections, layer norms, etc.

If we wanted a more quantitatively faithful “influence”, the principled object is the token-to-token Jacobian of the attention block at the input we care about:

$$J_{\text{attn}}^\ell = \left. \frac{\partial h^\ell}{\partial h^{\ell+1}} \right|_{\text{attn branch only}}$$

and then

$$\frac{\partial h^L}{\partial h^0} = \prod_{\ell=1}^L (I + J_{\text{attn}}^\ell), \quad A_{\text{eff}} = \frac{\partial h^L}{\partial h^0} - I$$

Attention sink normalisation

To address the phenomenon of attention sinks (tokens that accumulate disproportionately high attention weights regardless of semantic relevance) we implement a statistical outlier detection approach. For each output token position, we extract its attention distribution over all preceding input tokens and compute the median attention value across the output sequence. We then apply a log transformation (with a small constant added to prevent numerical instability: $\log(\text{attention} + \text{min}/2)$) to better capture the heavy-tailed distribution of attention weights. Tokens are identified as attention sinks when their log-transformed attention exceeds $\mu + 3\sigma$, where μ and σ are the mean and standard deviation of the log-attention distribution. These identified sink tokens can then be masked or down-weighted during attribution analysis. This approach effectively identifies grammatical tokens and separators that accumulate excessive attention (typically punctuation, articles, and structural tokens).

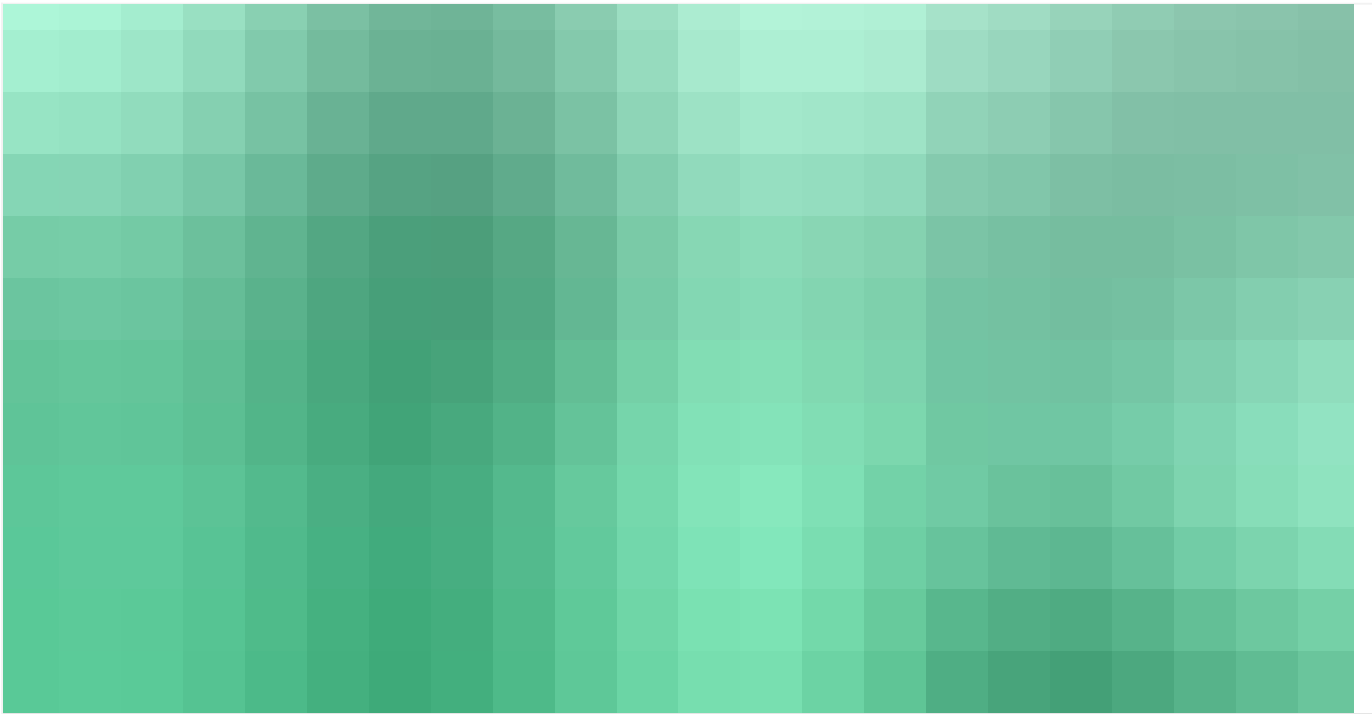
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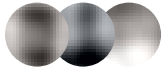
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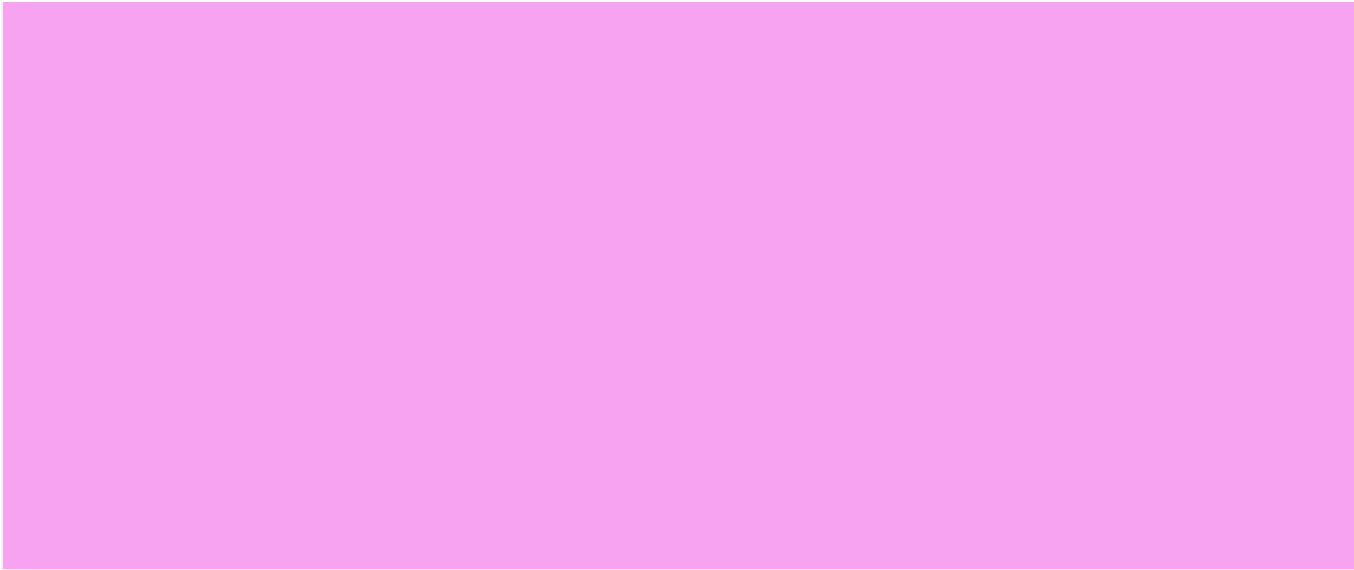
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